

Decision Control instructions -

Until now all programs are
used to execute sequentially.

1 print("Hello world") → "Hello world"

2 a = 10

3 b = 20

4 c = a + b

5 print(c) → 30

6 print("Program executed") → Program executed

```
graph TD; L1["1 print('Hello world') -> 'Hello world'"] --> L6["6 print('Program executed') -> Program executed"]; L2["2 a = 10"]; L3["3 b = 20"]; L4["4 c = a + b"]; L5["5 print(c) -> 30"];
```

We can change the sequence of execution by using

Decision control instructions.

- 1) if condition
- 2) if - else condition
- 3) if - elif - else condition

There are three ways we can make decision in our program.

if
(1)

if Condition:

↔ Stat 1
space Stat 2

if-else
(2)

if Condition:

↔ Stat 1
space Stat 2

else:

Stat 1

Stat 2

if-elif-else
(3)

if Condition:

Stat 1

Stat 2

elif Condition:

Stat 1

Stat 2

elif Condition:

Stat 1

Stat 2

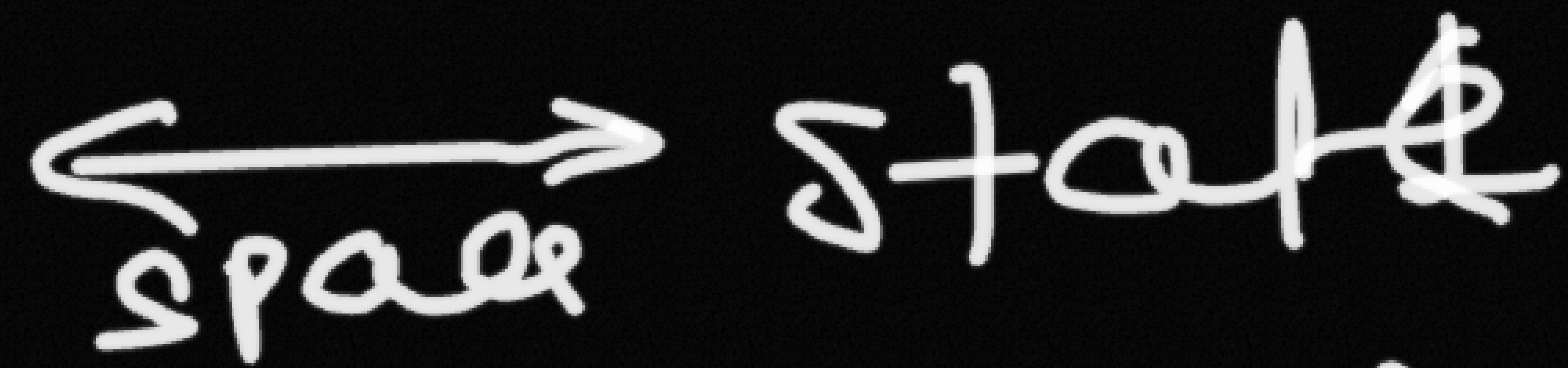
else:

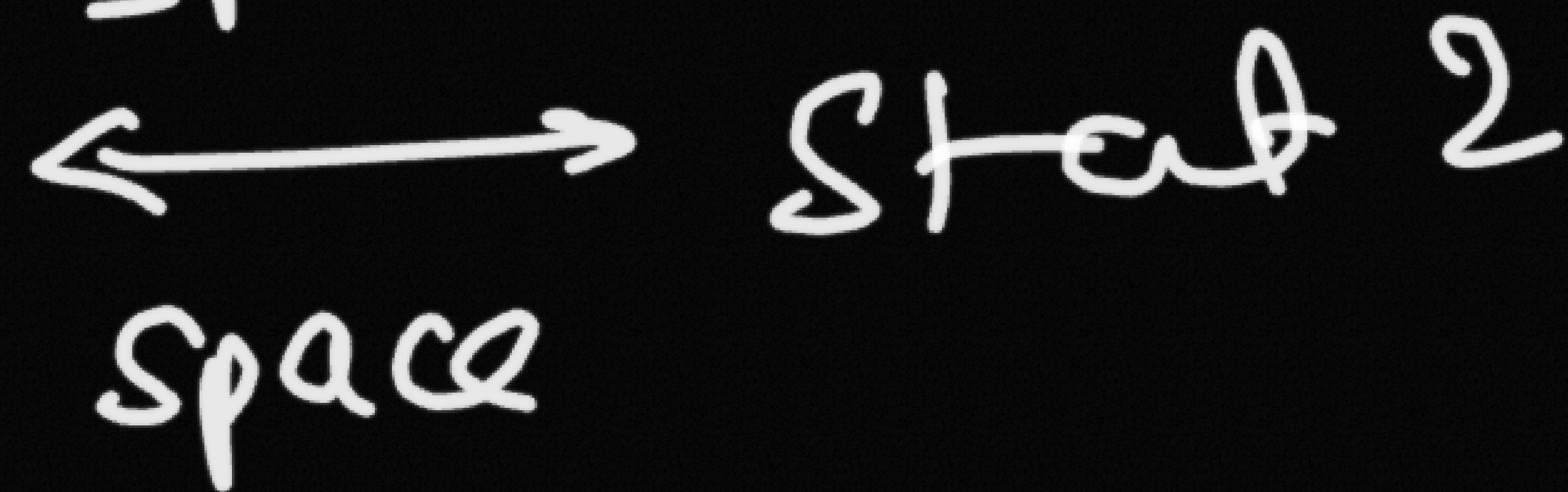
Stat 1
Stat 2

1) if block -

In if block, else block is optional

Syntax: if condition :

 stat 1

 stat 2

ex.

```
age = int(input("Enter your age"))
```

```
print(age)
```

condition (True)

```
if age ≥ 18:
```

```
    print("you are eligible for voting")
```

```
print("program executed successfully")
```

if block - if the if-block condition is true then all the statements inside the if-block will execute.

if the if-block condition is false then it will not execute any statement inside the if-block.

23 if-else block -

if condition :

Stat 1

Stat 2

else :

Stat 1

Stat 2

- if the if-block condition is true then all the statements inside the if block will execute.
- if the if-block condition is False then all the statements inside the else block will execute.

```
age = int(input("Enter your age:"))
```

```
print(age)
```

```
if age >= 18:
```

```
    print("you are eligible for voting")
```

```
else:
```

```
    print("you are not eligible for  
voting")
```

$<$ - less than

$<=$ - less than equal to

$>$ - greater than

$>=$ - greater than equal to

$==$ - equal to

$!=$ - not equal to.

3} if-elif-else block -

if condition:

Stat1

X

elif condition:

Stat2

X

elif condition:

Stat3

X

else:

Stat4

✓

- if the if-block condition is true then the statements inside the if block will execute.

- if the if-block condition is False then it checks the elif condition, if elif condition is true then the statements inside the elif block will execute.

- if first elif-block condition is False then it checks for next elif-block & if condition is true, it will execute statement inside 2nd elif block.
- & if 2nd elif-block condition is False, then statements inside the else block will execute.

age = int(input("Enter your age"))

~~3~~ ~~15~~ 25

~~1-9~~

10-17

10-17

18 →

17 →

if age < 10:

print("you are child")

elif age < 18:

print("you are teenager, but
not eligible for voting")

else:

print("you are eligible for voting")

